

0.1 重积分计算

定理 0.1 (二重积分的 Newton-Leibniz 公式)

设 $D = [a, b] \times [c, d] \subset \mathbb{R}^2$ 是闭矩形区域, 函数 $f(x, y)$ 在 D 上连续. 若存在二元函数 $F(x, y)$ 在 D 上具有连续混合偏导数 $\frac{\partial^2 F}{\partial x \partial y}$, 且满足

$$\frac{\partial^2 F}{\partial x \partial y}(x, y) = f(x, y), \quad \forall (x, y) \in D,$$

则

$$\iint_D f(x, y) dx dy = F(b, d) - F(b, c) - F(a, d) + F(a, c).$$



定理 0.2 (平面曲线积分的分部积分法)

设 L 是平面上的有向光滑曲线, 起点为 A , 终点为 B . 函数 $u = u(x, y)$ 与 $v = v(x, y)$ 在 L 上具有一阶连续偏导数. 则

$$\int_L u dv = uv \Big|_A^B - \int_L v du,$$

其中 $dv = v_x dx + v_y dy, du = u_x dx + u_y dy$, 且

$$uv \Big|_A^B = u(B)v(B) - u(A)v(A).$$



例题 0.1 设 $f(x, y)$ 在 D 上有定义, 且在某一圆盘外为 0. 求证:

$$\iint_D \left[\frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 \right] dx dy = 0.$$

证明 Show/Hide Proof

由题设可知 f 在 ∂D 上恒为 0, 进而

$$\frac{\partial f}{\partial x}(x, y) = \frac{\partial f}{\partial y}(x, y) = 0, \quad \forall (x, y) \in \partial D.$$

由 Green 公式可得

$$\begin{aligned} \iint_D \left[\frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 \right] dx dy &= \iint_D (f_{xx} f_{yy} - f_{xy}^2) dx dy \\ &= \iint_D [(f_{xx} f_y)_y - (f_{xy} f_y)_x] dx dy \stackrel{\text{Green 公式}}{=} \oint_{\partial D} f_{xy} f_y dx + f_{xx} f_y dy = 0. \end{aligned}$$

□

例题 0.2 设不全为 0 的实数 α, β, γ , 考虑

$$C: \begin{cases} x^2 + y^2 + z^2 = 1 \\ \alpha x + \beta y + \gamma z = 0 \end{cases}.$$

计算 $\int_C y(y-x) ds$.

解 Show/Hide Solution

记 $K \triangleq \sqrt{\alpha^2 + \beta^2}, M \triangleq \sqrt{\alpha^2 + \beta^2 + \gamma^2}$, 考虑如下正交变换

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \frac{\alpha\gamma}{KM} & \frac{\beta\gamma}{KM} & -\frac{K}{M} \\ \frac{\beta}{K} & \frac{\alpha}{K} & 0 \\ \frac{\alpha}{M} & \frac{\beta}{M} & \frac{\gamma}{M} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = T \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

于是曲线 C 方程等价于

$$\begin{cases} \left(\frac{\alpha\gamma}{KM}x' - \frac{\beta}{K}y'\right)^2 + \left(\frac{\beta\gamma}{KM}x' + \frac{\alpha}{K}y'\right)^2 + \left(-\frac{K}{M}x'\right)^2 = 1 \\ z' = 0 \end{cases} \iff \begin{cases} x'^2 + y'^2 = 1 \\ z' = 0 \end{cases}.$$

因为正交变换是等距变换, 进而保持弧长微元, 即

$$ds = \sqrt{(dx)^2 + (dy)^2 + (dz)^2} = \left| \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} \right| = \left| T \begin{pmatrix} dx' \\ dy' \\ dz' \end{pmatrix} \right| = \left| \begin{pmatrix} dx' \\ dy' \\ dz' \end{pmatrix} \right| = \sqrt{(dx')^2 + (dy')^2 + (dz')^2} = ds'.$$

又由曲线 $C: \begin{cases} x'^2 + y'^2 = 1 \\ z' = 0 \end{cases}$ 的对称性知

$$\int_{x'^2+y'^2=1, z'=0} x'y' ds' = \int_{x'^2+y'^2=1, z'=0} (-x')y' ds' \implies \int_{x'^2+y'^2=1, z'=0} x'y' ds' = 0.$$

所以

$$\begin{aligned} \int_C y(y-x) ds &= \int_{x'^2+y'^2=1, z'=0} \left(\frac{\beta\gamma}{KM}x' + \frac{\alpha}{K}y'\right) \left(\frac{(\beta-\alpha)\gamma}{KM}x' + \frac{\alpha+\beta}{K}y'\right) ds' \\ &= \int_{x'^2+y'^2=1, z'=0} \left[\frac{\beta(\beta-\alpha)\gamma^2}{K^2 M^2} x'^2 + \frac{\alpha(\alpha+\beta)}{K^2} y'^2 \right] ds' \\ &= \int_0^{2\pi} \left[\frac{\beta(\beta-\alpha)\gamma^2}{K^2 M^2} \cos^2 \theta + \frac{\alpha(\alpha+\beta)}{K^2} \sin^2 \theta \right] d\theta \\ &= \frac{\beta(\beta-\alpha)\gamma^2}{K^2 M^2} \pi + \frac{\alpha(\alpha+\beta)}{K^2} \pi = \pi \frac{\alpha^2 + \alpha\beta + \gamma^2}{\alpha^2 + \beta^2 + \gamma^2}. \end{aligned}$$

□

例题 0.3 对 $t \in (0, +\infty)$, 令

$$I(t) = \iint_{\Sigma_t} P dy dz + Q dz dx + R dx dy,$$

其中 Σ_t 是柱体 $\{(x, y, z) \mid x^2 + y^2 \leq t^2, 0 \leq z \leq 1\}$ 的表面, 定向为外侧. $P = Q = R = f((x^2 + y^2)z)$, $f \in C^1(\mathbb{R})$. 求极限:

$$\lim_{t \rightarrow 0^+} \frac{I(t)}{t^4}.$$

解 Show/Hide Solution

由 Gauss 公式可得

$$I(t) = \iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx dy dz = \iiint_V f'((x^2 + y^2)z) (2xz + 2yz + x^2 + y^2) dx dy dz,$$

其中 $V \triangleq \{(x, y, z) \mid x^2 + y^2 \leq t^2, 0 \leq z \leq 1\}$. 再利用柱坐标换元得

$$\begin{aligned} I(t) &= \iint_{x^2+y^2 \leq t^2} dx dy \int_0^1 f'((x^2 + y^2)z) (2xz + 2yz + x^2 + y^2) dz \\ &= \int_0^t dr \int_0^1 dz \int_0^{2\pi} f'(r^2 z) (2zr \cos \theta + 2zr \sin \theta + r^2) \cdot r d\theta \\ &= \int_0^t dr \int_0^1 2\pi f'(r^2 z) \cdot r^3 dz = \int_0^t 2\pi r f(r^2) dr. \end{aligned}$$

从而再由 L'Hospital 法则得

$$\lim_{t \rightarrow 0^+} \frac{I(t)}{t^4} = \lim_{t \rightarrow 0^+} \frac{I'(t)}{4t^3} = \lim_{t \rightarrow 0^+} \frac{2\pi t f(t^2)}{4t^3} = \lim_{t \rightarrow 0^+} \frac{\pi f(t^2)}{2t^2} = \frac{\pi}{2} f'(0).$$

□